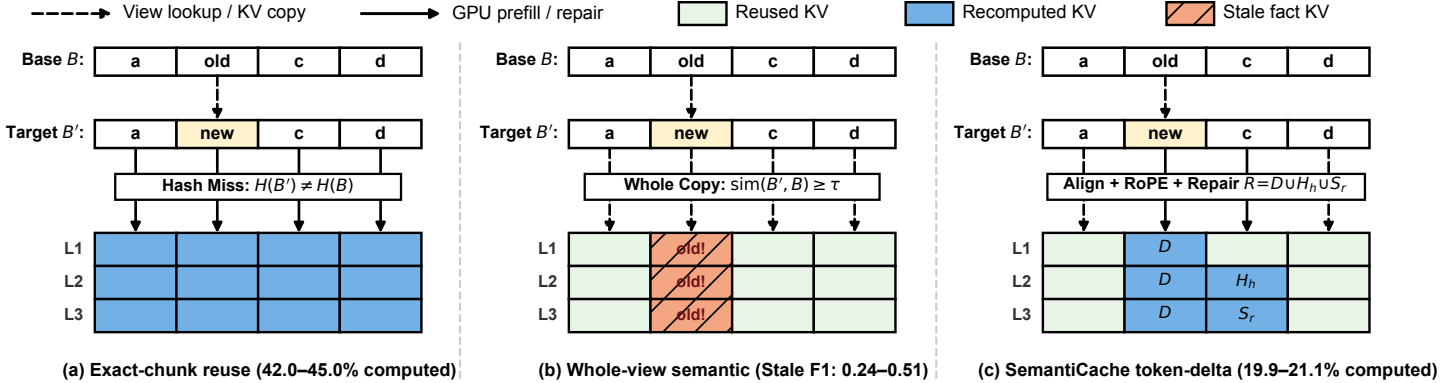
View lookup / KV copy
GPU prefill / repair
Reused KV
Recomputed KV
Stale fact KV
Base B:
a
old
c
d
Target B':
a
new
c
d
Hash Miss: $H(B') \neq H(B)$
L1
L2
L3
(a) Exact-chunk reuse (42.0–45.0% computed)
Base B:
a
old
c
d
Target B':
a
new
c
d
Whole Copy: $\mathrm{sim}(B', B) \geq \tau$
L1
L2
L3
old!
old!
old!
(b) Whole-view semantic (Stale F1: 0.24–0.51)
Base B:
a
old
c
d
Target B':
a
new
c
d
Align + RoPE + Repair $R = D \cup H_h \cup S_r$
L1
L2
L3
D
D
H_h
D
S_r
(c) SemantiCache token-delta (19.9–21.1% computed)